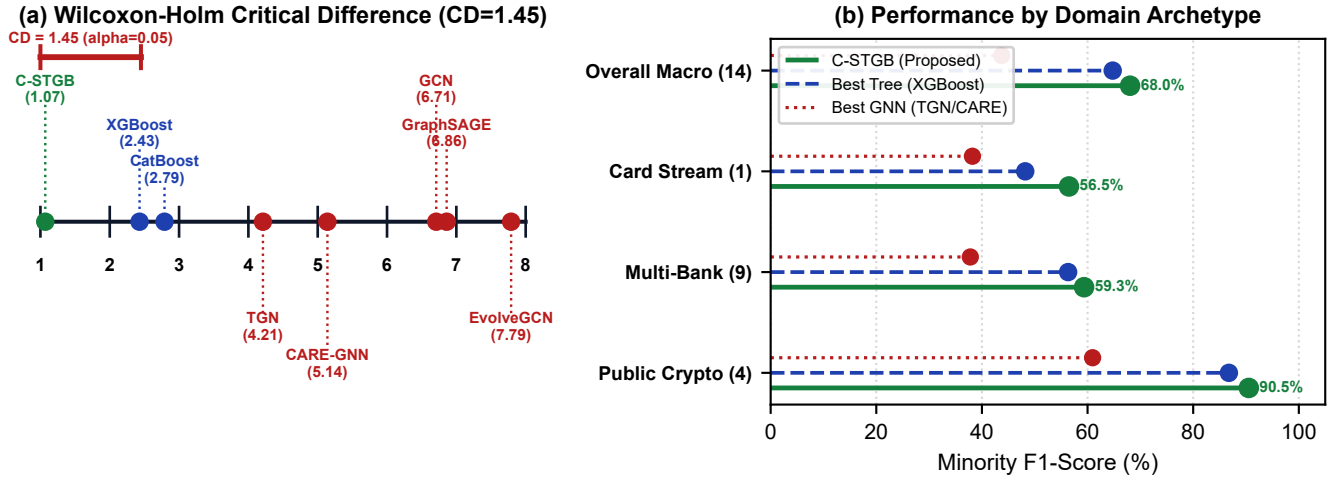

(a) Wilcoxon-Holm Critical Difference (CD=1.45)
CD = 1.45 (alpha=0.05)
C-STGB (1.07)
XGBoost (2.43)
CatBoost (2.79)
GCN (6.71)
GraphSAGE (6.86)
TGN (4.21)
CARE-GNN (5.14)
EvolveGCN (7.79)
1
2
3
4
5
6
7
8
(b) Performance by Domain Archetype
C-STGB (Proposed)
Best Tree (XGBoost)
Best GNN (TGN/CARE)
Overall Macro (14)
68.0%
Card Stream (1)
56.5%
Multi-Bank (9)
59.3%
Public Crypto (4)
90.5%
0
20
40
60
80
100
Minority F1-Score (%)